

Beyond Boundaries — verdict

Do the Xenium Multimodal Cell Segmentation images add anything beyond segmentation? Mouse EAE spinal cord, RRMAP2 runs 1–6: 54 images, 67 animals, 1.38 M cells. Status 8 October 2026; details in [BeyondBoundaries_methods_results.pdf](#) (methods + all numbers) and [BeyondBoundaries_lesion_story.pdf](#) (biology).

A modest yes. The images are first of all a segmentation aid. Beyond that they give a small number of robust readouts the 5K panel cannot. Their pooled channels and optical bleed limit them as a protein assay, so specific protein questions need dedicated stains.

What holds (replicated across runs, with controls)

Readout	Evidence	Use
Segmentation (the kit's purpose)	18S interior stain draws ~95 % of cell masks	real; not benchmarked against alternatives here
White-matter neuropil loss (ATP1A1 around each cell; <i>Atp1a1</i> not on the panel)	lesion WM $\times 0.79$ (runs 5/6) and $\times 0.72$ (runs 1–3, $p < 10^{-4}$) vs healthy WM of the same piece	tissue damage the RNA can't show; weaker with stricter lesion calls ($\times 0.93$); crowding control still to do
Astrocyte reactive RNA comes before vimentin protein (<i>Vim</i> not on the panel)	protein vs RNA p 0.46–0.58 per run; "RNA-only" astrocytes enriched in active disease in both run sets	the clearest protein-vs-RNA insight
Images alone map lesions and cell types on unseen runs	lesion AUROC 0.85–0.91, cell type 0.42–0.47 (chance 0.07), leave-one-run-out over 5 runs	QC and annotation cross-check; redundant with RNA for biology

What is weak or partial

- Lesion vimentin and outcome:** low at peak, fills lesion interiors, tends to be higher in animals that do well ($p \approx -0.45$ after time adjustment). But part of it is shared with VSMCs in the same channel (astrocyte-specific $p \approx -0.33$), it is entangled with disease burden, and the same-image test is weak (12/19 pairs). **Weak to moderate.**
- 18S texture as a severity marker:** replicates in astrocytes, myeloid cells and oligodendrocytes only.
- 18S gain in activated lesion cells beyond RNA content:** plausible, not stress-tested.

What does not work, or was withdrawn after controls

- CD45** is undetectable in mouse spinal cord; the boundary channel reports ATP1A1 only.
- No hidden cell states:** ~85 % of image variance isn't explained by RNA, but it is mostly a local optical/staining field.
- Images don't improve lesion identity or severity prediction** over the transcriptome.
- Withdrawn:** T cells orienting 18S towards vessels (bleed from bright neighbours + outline geometry); a vimentin ring at lesion edges and vimentin-border containment (artefacts of a lesion-object definition). No evidence that vimentin is a barrier to immune cells (within-lesion test $p +0.035$, 95 % CI -0.016 to $+0.073$).

Why: three structural limits

- Pooled channels** (α SMA + vimentin; ATP1A1 + CD45 + E-cadherin): every protein claim carries "or the other one".
- Optical bleed and outline geometry:** small cells next to bright neighbours borrow their signal; polarity and "around the cell" measures need brightness-matched and nucleus-relative controls.
- The stains were chosen to draw cell boundaries**, not to answer biological questions.

Recommendations

- Keep the kit** for segmentation; use neuropil loss and astrocyte vimentin state as add-on readouts, with the caveats above.
- For protein biology:** post-Xenium immunofluorescence on the same sections with single-target antibodies (vimentin, α SMA, GFAP; a mouse-reactive CD45). The most informative sections: MILD16 vs SEVERE16 (they share images) and MONOPHASIC vs REMISSION1.
- Future panels:** add *Vim*, *Acta2*, *Gfap*, *Atp1a1* as custom genes.
- Worth writing up as methods:** control-referenced lesion definitions; lesion regions instead of linked cells; brightness-matched and nucleus-relative controls for polarity.

The bigger contribution may be the disease biology (transcriptome-led, all five runs)

Lesions defined against controls (controls ~1 %, not 4–20 %) resolve along a clear course: active monocyte-derived infiltrate → fibrotic, demyelinated and glial-reactive tissue. Chronic severity is set at the first attack and means lesions that stayed active. Relapse can't be predicted clinically; B cells accumulate after the first attack in meningeal aggregates (fewer in never-relapsing animals), and relapses are mostly new lesions.

For discussion: (1) the paper's framing, disease biology or kit evaluation; (2) whether the slides are available for post-Xenium immunofluorescence; (3) which open threads to close first (neuropil-loss crowding control; B cells and relapse with more REMISSION1-stage animals).